

E-Commerce Management System

Module: Customer Management Module — End User & Technical Documentation

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GitHub Repo: ansarizeenat/ecommerce-category-management	Live Server (Render): https://ecommerce-category-management.onrender.com
Tech Stack: Java 25, Spring Boot 4.1, JPA, MySQL 8.0, Thymeleaf	Date: September 2026

1. Introduction

The **Customer Management Module** manages customer data, including personal contact information, order history, and account status settings. It enables customers to register directly through the frontend, allows administrators to onboard and modify customer profiles, inspect their lifetime purchase history and orders, and safely deactivate customer accounts via a non-destructive soft-delete mechanism.

2. Database Design & Specification

The database schema strictly conforms to the Itvedant design specification for the users table:

Column Name	Data Type	Key / Constraint	Description
user_id	INT / BIGINT	PK, Auto-Increment	Unique identifier for each customer
first_name	VARCHAR(100)	NOT NULL	Customer's first name
last_name	VARCHAR(100)	NULLABLE	Customer's last name
email	VARCHAR(100)	NOT NULL, UNIQUE	Customer's unique email address
phone	VARCHAR(15)	NULLABLE	Customer's contact phone number
created_at	DATETIME	AUTO-GENERATED	Timestamp when customer account was created
updated_at	DATETIME	AUTO-UPDATED	Timestamp when customer details were last updated
status	BOOLEAN	NOT NULL, DEFAULT TRUE	True for active customers, False for inactive (soft delete)

3. Core Functionalities & Business Logic

1. Add a New Customer: Customers register through the web interface or administrators add customer details manually. The system enforces mandatory fields (First Name, Email) and unique email validation to eliminate duplicate accounts.

2. Customer Dashboard: Administrators view all customers, contact details, account status, lifetime spend, and total orders placed. Quick status filters (All, Active, Inactive) streamline account management. An integrated **Order History Modal** displays every order placed by the selected customer.

3. Update Customer Details: Administrators can modify customer names, email addresses, phone numbers, and status. Email collision checks ensure integrity, and the `updated_at` timestamp is recorded.

4. Delete / Deactivate Customer Account (Soft Delete): Administrators can deactivate accounts without removing data. Sets `status = false` while retaining historical orders and metrics. Deactivated accounts can be reactivated with one click.

4. REST API Endpoint Specifications

Method	Endpoint	Description	Sample Payload / Response
POST	<code>/api/customers</code>	Add / Register customer	<code>{"firstName": "Zeenat", "lastName": "Ansari", "email": "zeenat@example.com", "phone": "+91 9876543210"}</code>
GET	<code>/api/customers</code>	Get all customers (supports <code>?status=true/false</code>)	Returns list of CustomerDTOs with total orders and lifetime spend
GET	<code>/api/customers/{id}</code>	Get customer profile by ID	Returns customer entity JSON
GET	<code>/api/customers/{id}/orders</code>	Get customer with full order history	Returns customer profile + array of orders placed
PUT	<code>/api/customers/{id}</code>	Update customer details	<code>{"firstName": "Zeenat", "email": "zeenat.new@example.com"}</code>
DELETE / PUT	<code>/api/customers/{id}</code> <code>/api/customers/{id}/deactivate</code>	Soft delete / Deactivate account	Sets <code>status=false</code> , updates timestamp
PUT	<code>/api/customers/{id}/activate</code>	Reactivate customer account	Sets <code>status=true</code> , updates timestamp

5. Verification & Automated Test Results

Automated unit and integration tests were implemented in `CustomerServiceTest.java`:

- **testCreateCustomer_Success:** Validated registration, default active status, and auto-timestamps.
- **testCreateCustomer_DuplicateEmailThrowsException:** Validated unique email constraint.
- **testUpdateCustomer_Success:** Validated customer profile editing.
- **testDeactivateCustomer_SoftDelete:** Validated soft delete behavior (`status=false`, record preserved).
- **testActivateCustomer_Reactivation:** Validated reactivation back to active status.
- **testGetCustomerWithOrders_OrderHistory:** Validated customer order history linkage.

Suite Test Summary: [INFO] Tests run: 13, Failures: 0, Errors: 0, Skipped: 0 — BUILD SUCCESS

6. Assignment Submission Summary

Item	Details & Links
1. GitHub Repository	https://github.com/ansarizeenat/ecommerce-category-management
2. Free Cloud Hosted URL	https://ecommerce-category-management.onrender.com
3. End User Documentation	Prepared in PDF format (attached document).
4. Modules Completed	1. Category Management 2. Product Management 3. Order Management 4. Customer Management